PalmSource™ 99

Wireless Opportunities from Motorola

Omid Tahernia
General Manager, Wireless Communications Div.
Motorola's Semiconductor Products Sector

3Com
More connected.

Introduction

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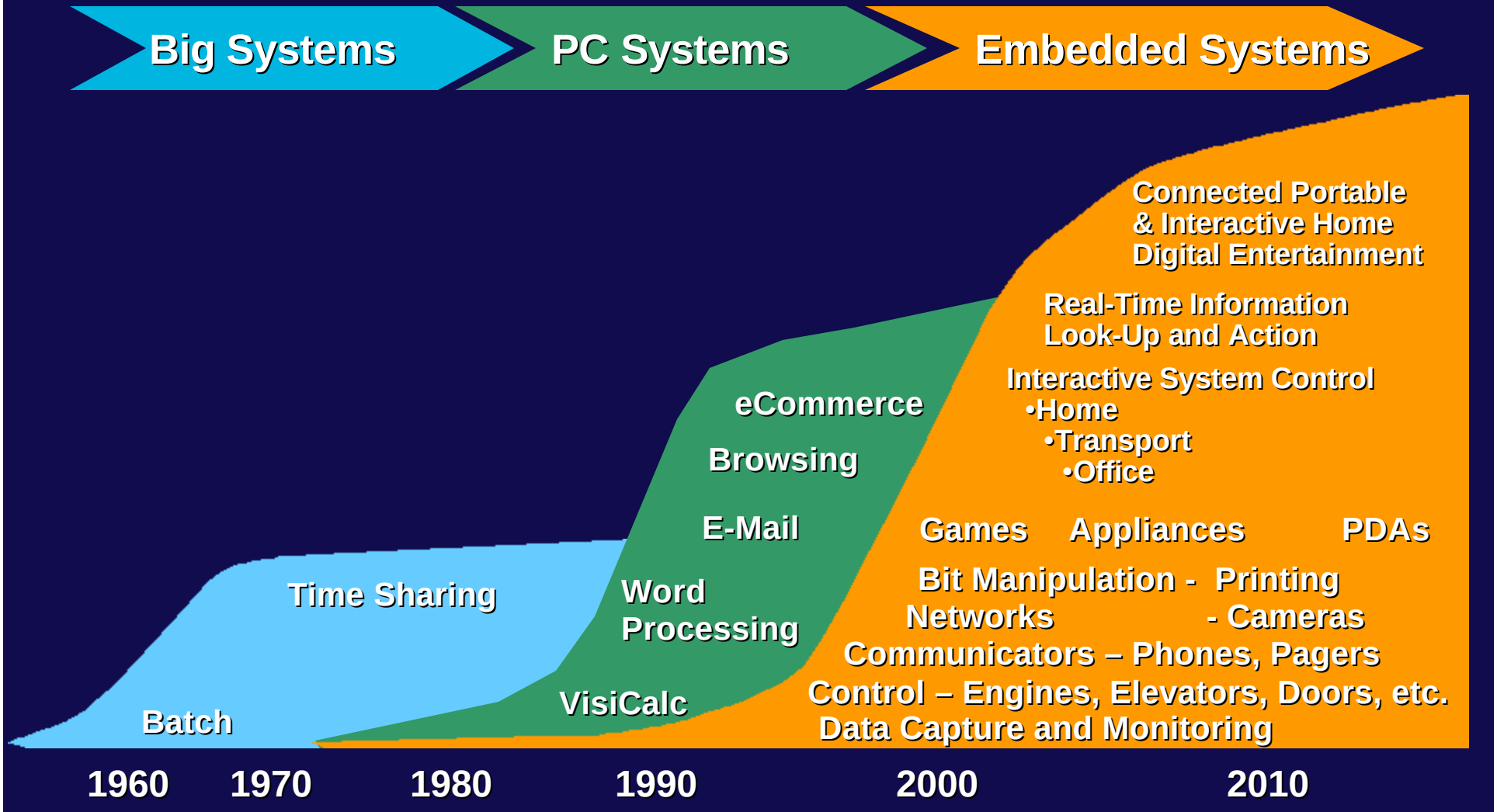
- Our portfolio ranges from Motorola-branded consumer products to integrated systems and embedded solutions that differentiate our customers' products

Communications Solutions	Embedded Semiconductor Systems	Integrated Electronic Systems
Software-enhanced wireless phones, 2-way radios, messaging and satellite communication products, networking and Internet access products	Embedded semiconductor solutions for customers in the networking, computing, imaging and entertainment, transportation and wireless communication markets	Electronic systems for automotive, communications, imaging, manufacturing systems, computer and consumer markets

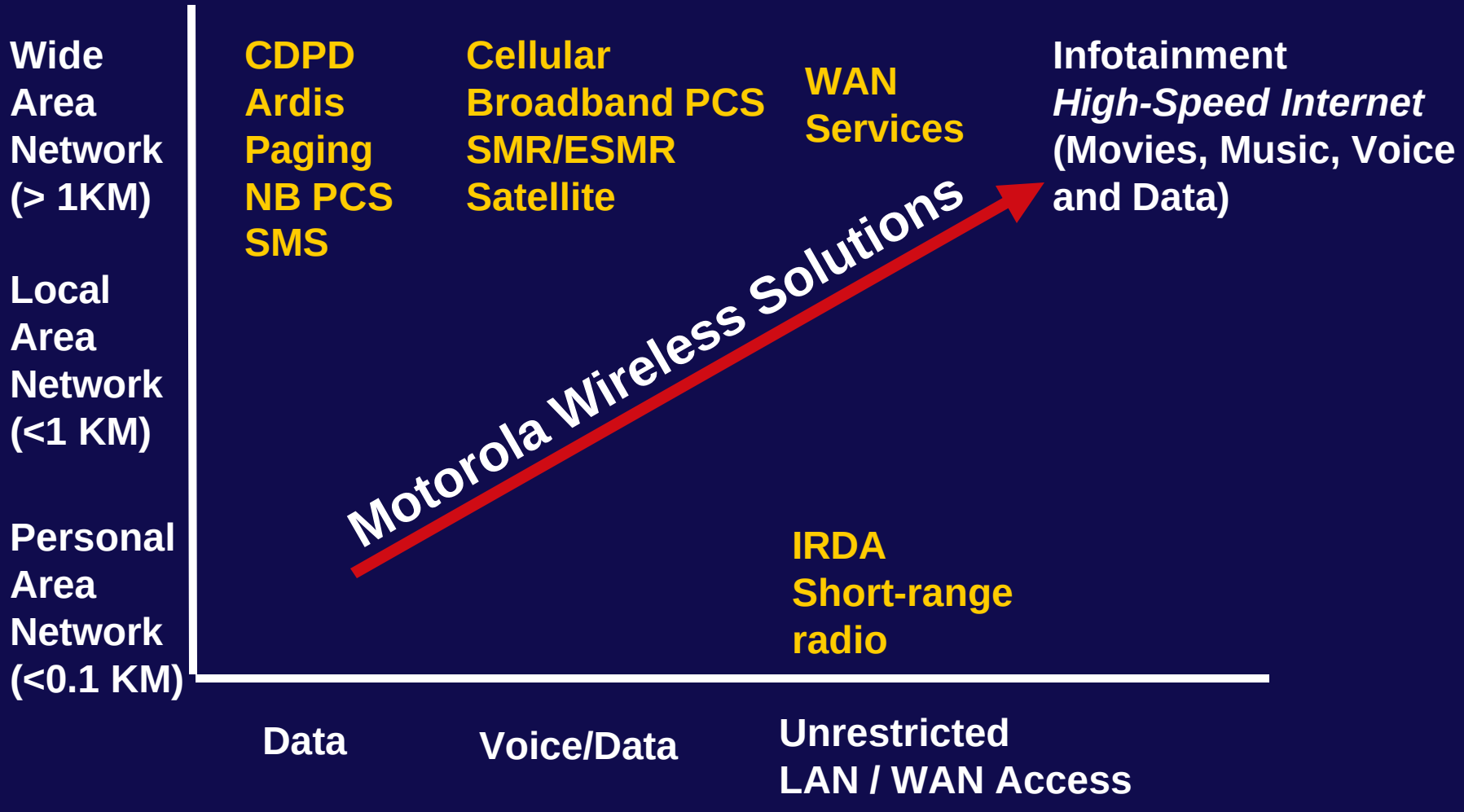
Digital DNA™ from Motorola

Third Wave of Electronic Computing

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Wireless Solutions Enabling the Future



Devices Converge to Access the Wireless Internet

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PC Companion
Classic PDA
Pen Notepad



Super PC Companion
Pen Tablet
Smart PDA





Wireless Everything:
PDAs, Tablets
Streaming Video
3G, MP3 Audio

Two-way Radio
Cellular Phone
Wireless Data
Paging



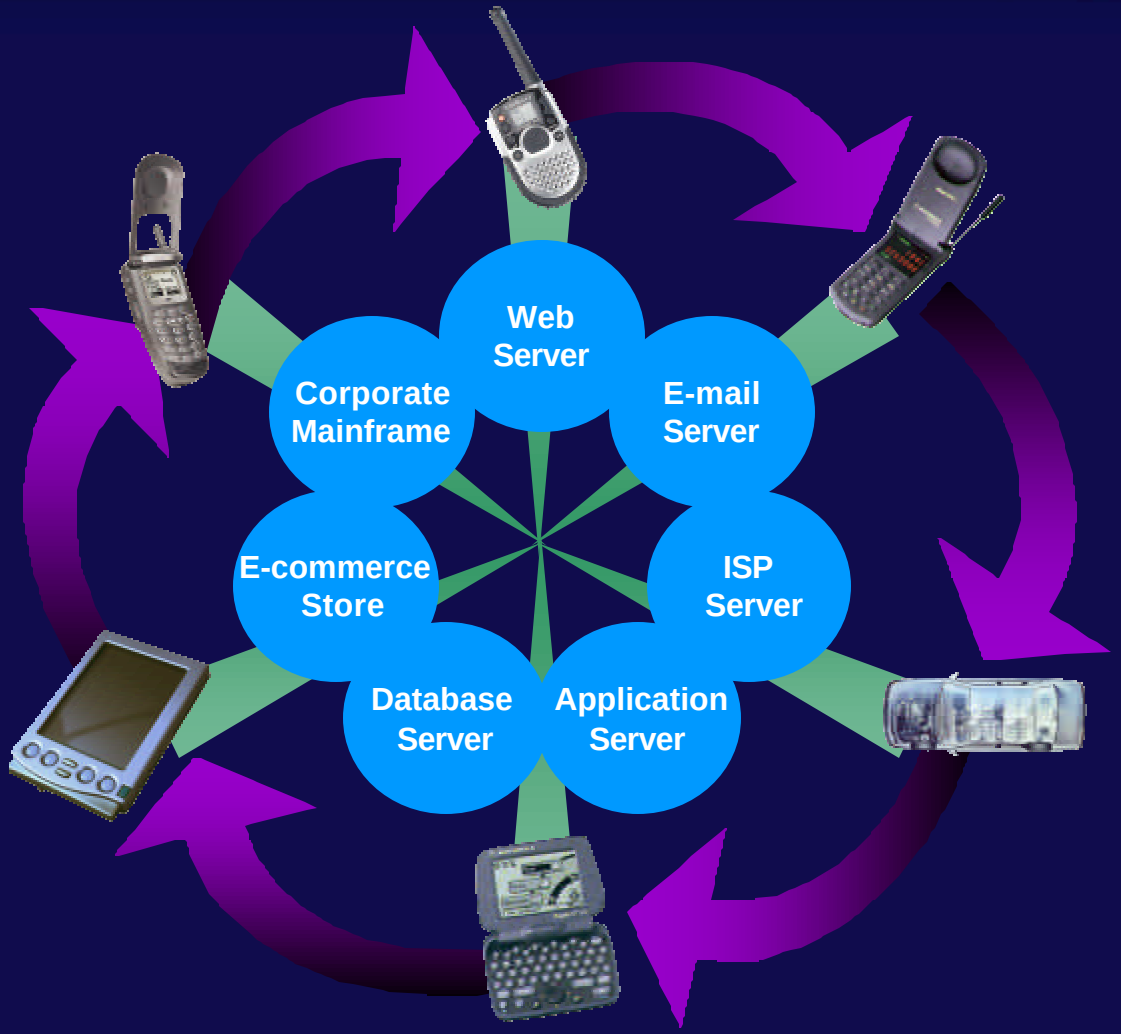
Smart Phones
Wireless Browsers





Thinking and Linking

- In wireless markets, the trend is to ubiquitous connectivity. Solutions are needed that not only THINK, but LINK to everything else



Building on Solid Foundations

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Emerging Market Differentiators

Solid Foundation

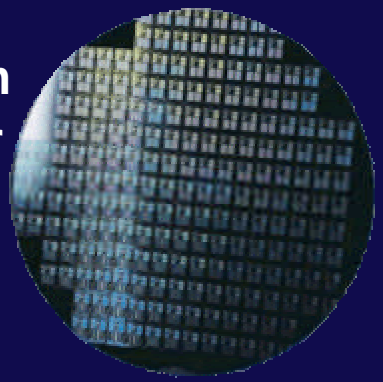
Wireless Value Propositions

- World Class Cores and System-On-Chip Methodology
- Leadership RF/IF and PA
- World Class Technology and Manufacturing
- Turn-Key Platforms
- Leadership in Smart Applications

Semiconductor Technology

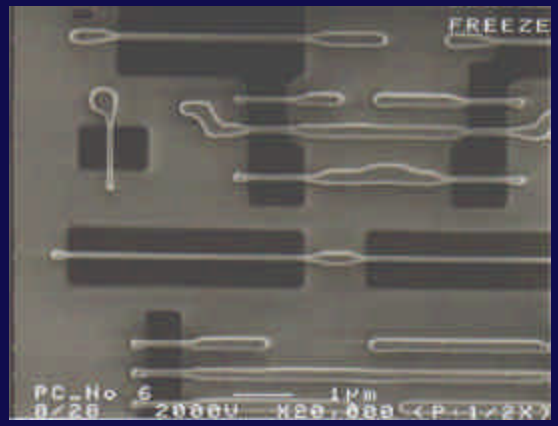
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300mm Wafer

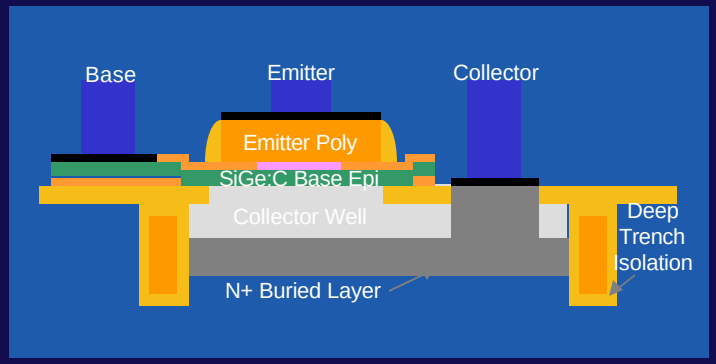


Dual Inlaid Copper Interconnect





0.1—Micron Lithography



Silicon Germanium Carbon

DragonBall™ Series Processors



Multi-Purpose Integrated Processor



Millions In Use

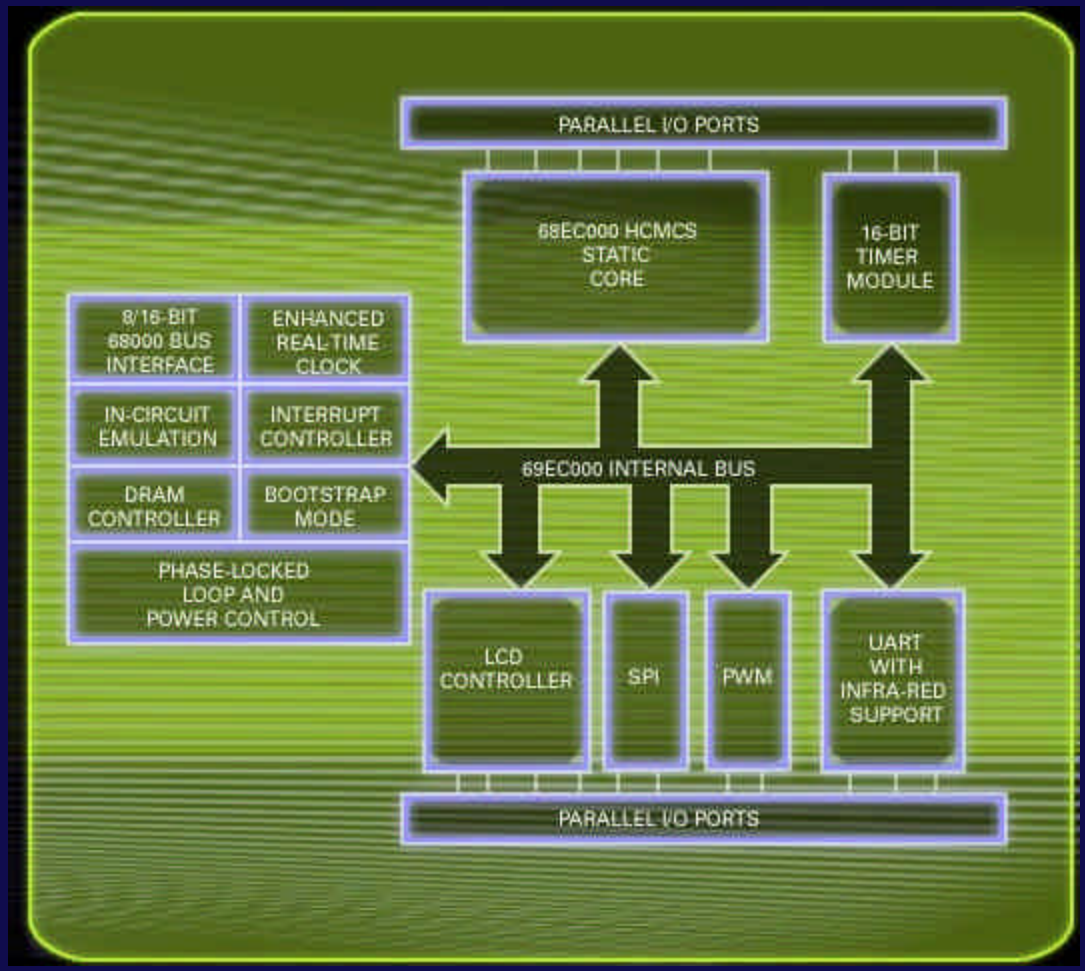
#1 Best Selling—PDA Processor

Featured in All Palm Pilot Organizers



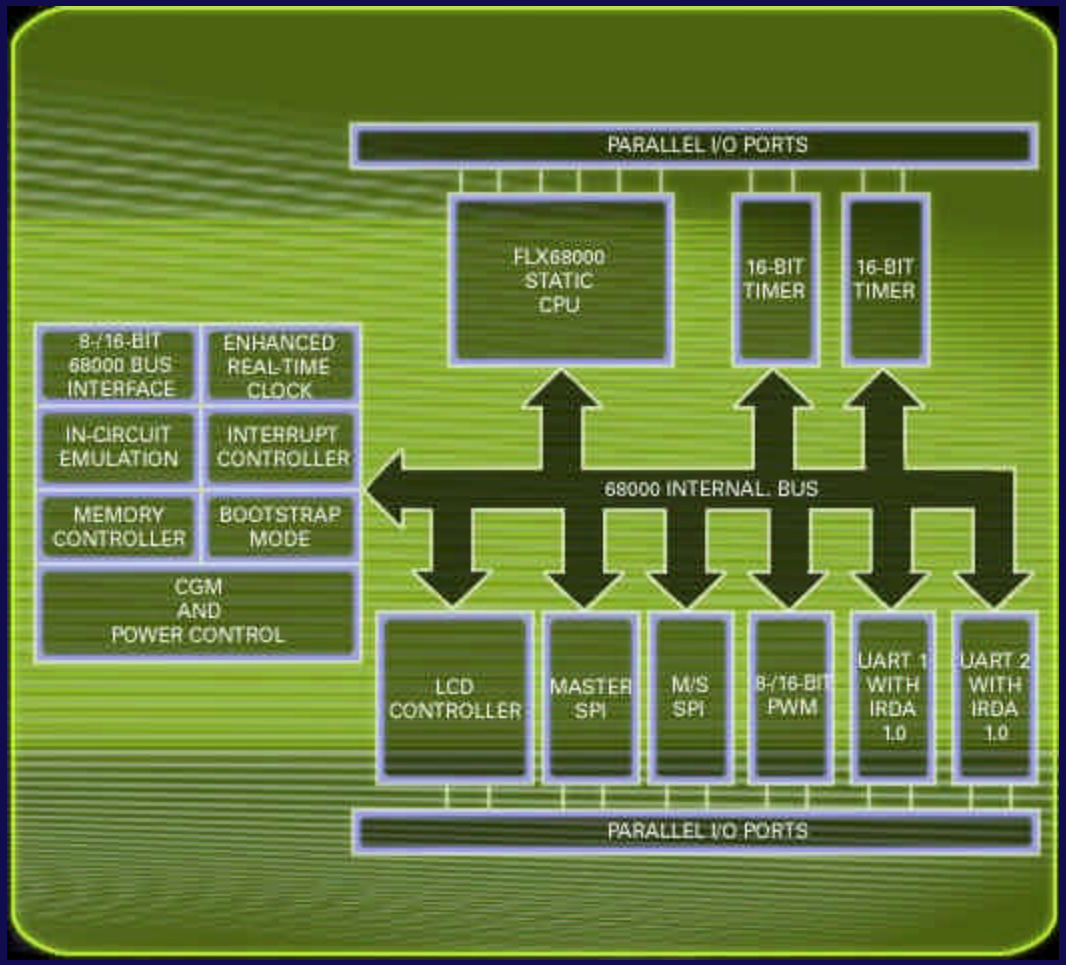
DragonBall EZ

- #1 Best Selling PDA Processor in the World
- Optimized for PDA applications
- Now available at 20MHz and 16MHz

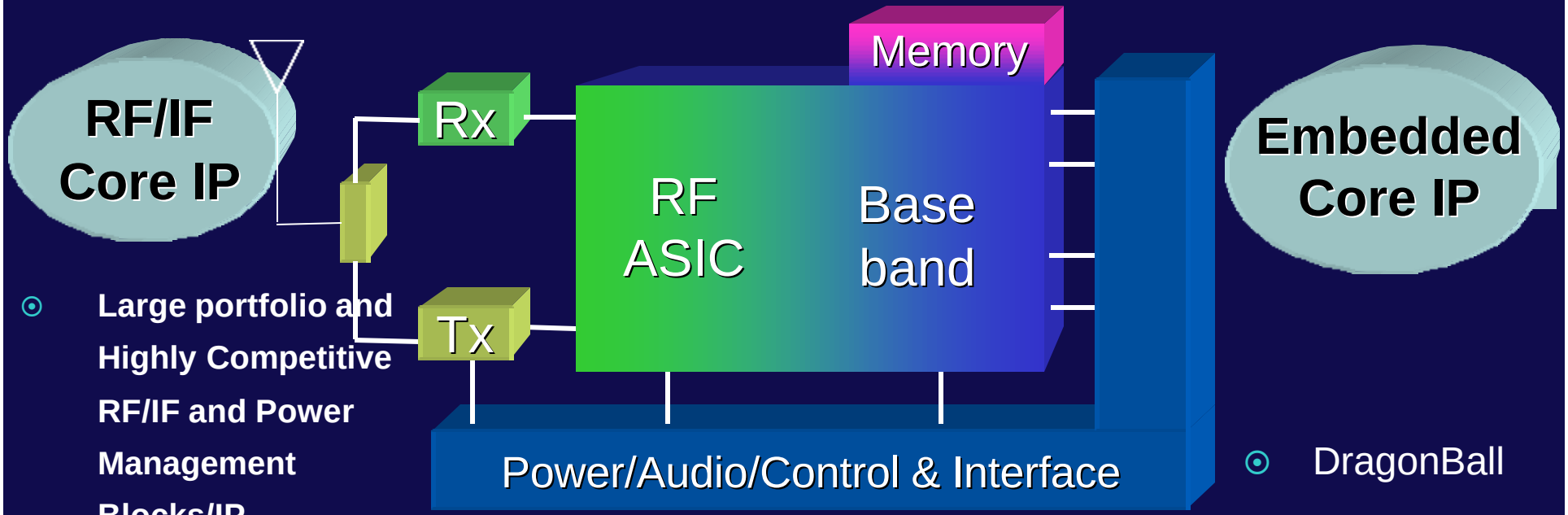


DragonBall VZ

- New DragonBall VZ Announced at PalmSource99!
  - 33MHz
  - Additional I/O pins
  - 2 SPIs
  - 1.2 Meg iRDA port
  - Synchronous DRAM controller



Total Wireless Solutions



- Large portfolio and Highly Competitive RF/IF and Power Management Blocks/IP
- Now supply over 10 major Merchant Market wireless manufacturers with RF/IF ASICS

- DragonBall
- DSP
- Peripherals

Motorola/Sun KJava™ Partnership

- Sun Microsystems has licensed Motorola to use and distribute the full family of Java platform technologies
- Motorola has incorporated EmbeddedJava™, PersonalJava™ and JavaCard™ technologies into its total solutions portfolio
- New Java Virtual Machine—KJava™
  - Motorola/Sun's co-development provides a time-to-market advantage
  - Announced at JavaOne in San Francisco, June 15, 1999
- KJava virtual machine
  - Optimized for small, portable devices
  - Allows content- and service-providers to easily support multiple platforms with "zero-footprint" applications
  - Allows network operators to distribute value-added software and upgrades over wireless (and wireline) infrastructures

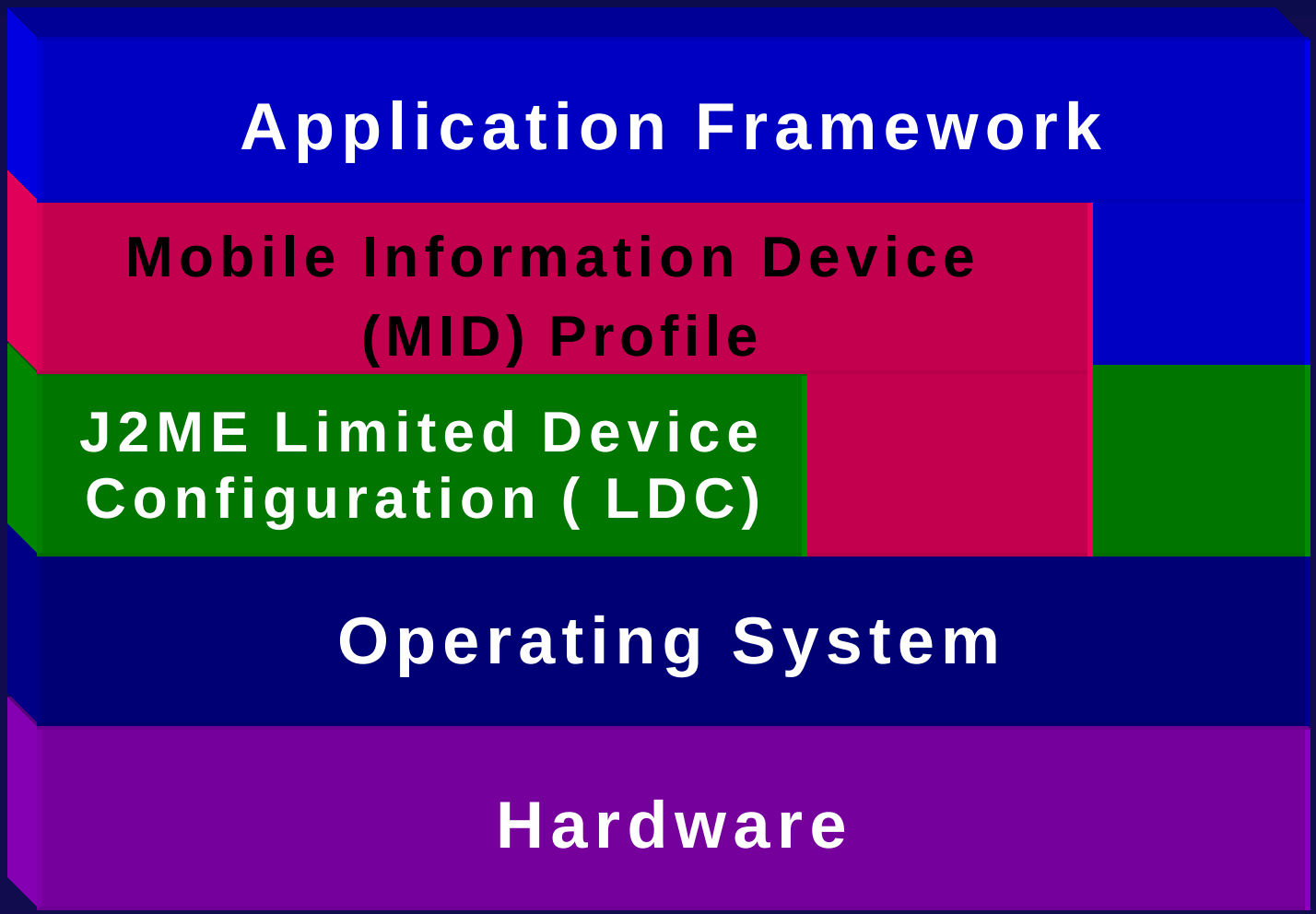


Java™ 2 Micro Edition (J2ME)

- Created by Motorola, Sun Microsystems, Groupe Bull and 3Com
- Includes a small sized virtual machine (KVM)
  - Specifically designed for resource-frugal devices where traditional Java was too large and slow
  - Initially targets mainly wireless devices
  - High-performance, Java-conformant environment
- Will eventually contain series of profiles
- Key wireless service providers and manufacturers supporting KVM technology worldwide
- Key KVM platform prototypes (incl. Motorola's) demo'd at JavaOne Developers Conf., June 1999

J2ME Wireless Device Architecture

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Java 2 Micro Edition

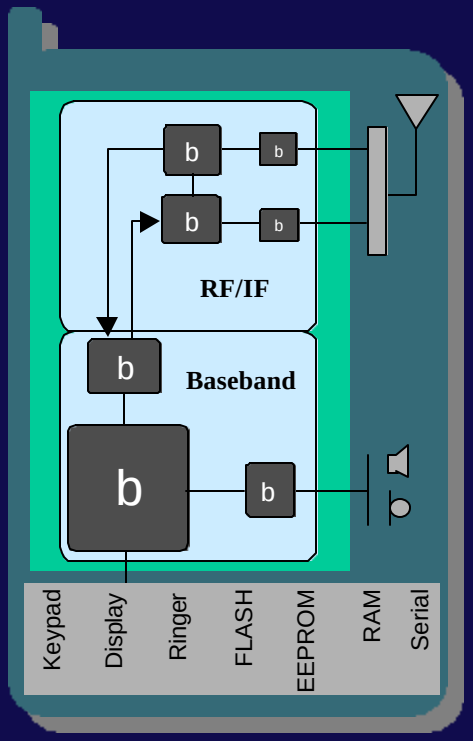
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Target Device Characteristics	*Limited Dev Config (LDC)*	*Mobile Info Device (MID)*
• *<256K ROM*	• *KVM*	• *Wireless telephony*
• *<256K RAM*	• *Core library*	• *Messaging*
• *3-5 MIPS*	• *UI, graphics*	• *Power monitoring*
• *9600 BPS*	• *Event model*	• *Calendar*
• *Battery powered*	• *Security model*	• *Phonebook*
• *Limited display*	• *Network*	
	• *Apps installation*	
	• *Persistent storage*	

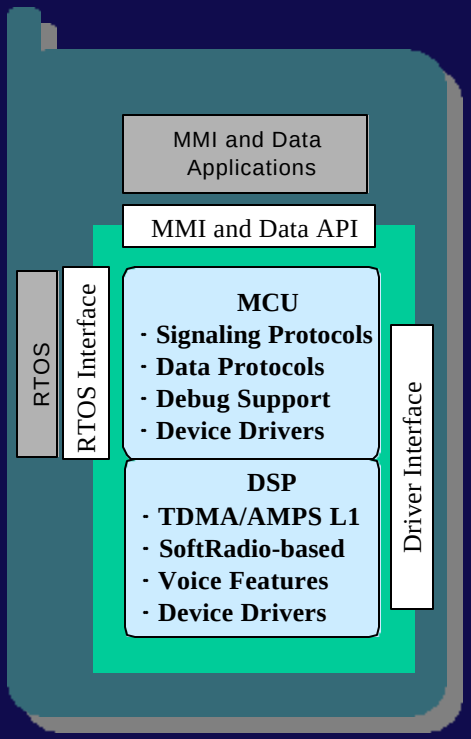
GSM/TDMA Platform

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Chipset
Baseband—RF/IF



Software
Integrated Layer 1–3



Development Platforms



Bluetooth Technology

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- Bluetooth uses a short-range radio link that enables the exchange of information between many devices, including mobile telephones, PDAs, notebook PCs, handheld PCs, associated peripherals, and home hubs, which may include Home RF
  - Radio operates on the globally available, 2.45 GHz ISM license-free band
- Bluetooth radio-link eliminates line of sight connection requirement. Bluetooth transactions can occur from products in pockets, handbags or briefcases
- Combination of circuit- and packet-switching, making it suitable for both voice and data

Bluetooth Scenario

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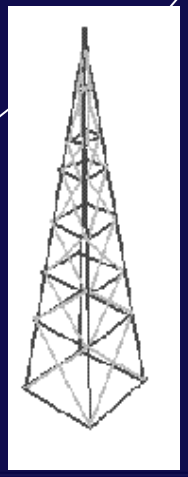
User makes a telephone number change on the Palm device



which is transmitted via Bluetooth to the PIM residing on the user's laptop in the briefcase… then to the PIM in the user's cellular phone. It is transmitted to the home network AND/OR the server in the work environment via the cellular network……







Cable

OR

PSTN

Home Environment





SNAP receives info via PSTN and transmits it via Powerline/ HomePNA/ Home RF or Bluetooth to the home computer

Work Environment

Corporate Server

Corporate server receives information via PSTN

Bluetooth Demo
In Solutions Pavilion

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Palm Computing and iDEN

- Opportunities for developer community to create 3rd Party Applications that bring the value of packet data and the iDEN product portfolio to the Palm platform
- Software Developer Kit (SDK) for iDEN/Palm applications now available and being demonstrated in the Solutions Pavilion



i1000plus

Wireless Standard Evolution

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	1G	2G				2.5G			3G	
Name	**AMPS**	**IS95A/B**	**GSM**	**IS136**	**PDC**	**IS95C 1X**	**GPRS**	**EDGE**	**IS95C 3X**	**WCDMA**
MA	FDMA	NBCDMA	TDMA	TDMA	TDMA	NBCDMA	TDMA	TDMA	WBCDMA	WBCDMA
kbps	0	14.4/64	14.4	14.4	9.6	144	128	384	384	384
									3GPP2	

- 3G is a wireless standard that meets or exceeds the IMT-2000 requirements set by the ITU
  - Increased data rates
    - 144 kbps @ mobile speeds
    - 384 kbps @ pedestrian speeds
    - 2 Mbps @ stationary environment
      - High speed Internet, video and location services

SnapTrack™ Enhanced GPS

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- Through software, SnapTrack momentarily turns a wireless device into a GPS receiver and sends your precise location with your call for help
- Tight integration of a GPS system with a wireless communications network…Client (handset) Server (a PC at MSC) Architecture
- Powerful, new GPS signal processing algorithms implemented in software

Performance vs. Conventional GPS

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	SnapTrack	Conventional GPS
Average Accuracy	3–35 m	50–200 m
Time to first fix	1 second	Up to several minutes
Fix rate	Excellent	Poor with any blockage
Power consumption	Minimal	High
Cost to deploy	Shared circuitry	Expensive

Summary

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- Embedded Wireless Computing:
  - The shift toward intelligent products that think and link
- Integration and Manufacturing expertise a must:
  - Copper connects, 300mm wafers, advanced RF, System-on-chip design methodology
- DragonBall continues to thrive in portable applications
  - New DragonBallVZ drop-in replacement that doubles performance
- Semiconductor Manufacturers must be able to add value in software:
  - SnapTrack, Bluetooth, Java, Cellular protocols, etc.


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More connected.